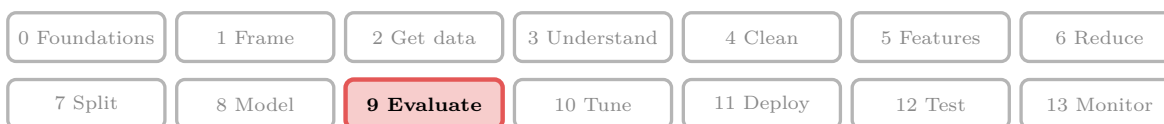


Classification Metrics: Accuracy and the Confusion Matrix

Contents

1. Overview	2
2. Accuracy	2
2.1 The idea	2
2.2 A real example	2
2.3 More than two classes	3
3. How much accuracy is good enough?	3
4. The confusion matrix	3
4.1 What accuracy hides	4
4.2 Reading it	4
4.3 The names	5
4.4 Type I and Type II errors	5
4.5 Accuracy from the confusion matrix	6
5. More than two classes	6
6. When accuracy misleads: imbalanced data	7
7. Summary	8
8. Key terms	8

Where this fits: Pipeline map step 9 (Evaluate). See the [Course map](#).



- **Builds on:** Imbalanced data ([Note 9](#)); Train-test split ([Note 13](#)).
- **Leads to:** Precision, recall and F1 ([Note 77](#)); ROC curve and AUC ([Note 78](#)).
- **Compare with:** Type I and II errors, power, tails ([Note 292](#)).

1. Overview

Key point: Accuracy is the fraction of correct predictions. The confusion matrix splits the predictions into four counts, which also show what kind of mistakes a model makes.

Regression models were judged with regression metrics such as R^2 (the regression metrics Note). Classification models need their own **classification metrics**. There are several: accuracy, the confusion matrix, precision, recall, F1 score, ROC-AUC and more.

This Note covers the first two. The following Notes build on the confusion matrix.

2. Accuracy

Key point: Accuracy = number of correct predictions / total number of predictions.

2.1 The idea

Key point: Compare each prediction with the true label and count how many match.

Suppose we train two models on the student placement data, for example logistic regression and a decision tree (a later Note), and predict the test students. For each test student we know the true result, so we can tick each prediction as right or wrong.

$$\text{accuracy} = \frac{\text{number of correct predictions}}{\text{total number of predictions}}$$

With numbers: if a model gets 8 of 10 test students right, its accuracy is $8/10 = 0.8$, or 80%. A model with 9 of 10 right (90%) is better on this test set.

2.2 A real example

Key point: On the heart-disease data, logistic regression reaches 0.869 and a decision tree 0.836.

The heart-disease dataset has 303 patients, 13 medical measurements each (age, blood pressure, cholesterol and so on), and a target: 1 if the patient has heart disease, 0 if not. We split off 20% as the test set (random state 2) and train both models.

Python: Accuracy in scikit-learn.

```
from sklearn.metrics import accuracy_score

lr = make_pipeline(StandardScaler(), LogisticRegression())
lr.fit(X_train, y_train)
accuracy_score(y_test, lr.predict(X_test))    # 0.869
```

Model	Test accuracy
Logistic regression (inputs standardised)	0.869
Decision tree	0.836

So, on this test set, logistic regression is the better model.

2.3 More than two classes

Key point: Accuracy is computed the same way for any number of classes.

Nothing changes with more classes, for example the three iris species: count the correct predictions and divide by the total. Logistic regression on the iris test set (30 flowers) gets 29 right: accuracy 0.967.

3. How much accuracy is good enough?

Key point: It depends on the problem. 99% can be far too low for cancer detection and plenty for predicting food orders.

A common interview question is “what accuracy should a model have?”. There is no single number; it depends on the cost of a mistake.

- **Cancer detection from X-rays:** 99% accuracy means one patient in a hundred gets a wrong answer. That is not acceptable.
- **Self-driving car decisions:** 99% means an error every hundred decisions. Also not acceptable.
- **Predicting whether a customer orders food this weekend:** 80% may be perfectly useful, because a wrong guess costs little.

4. The confusion matrix

Key point: A table of actual classes against predicted classes. For two classes it has four counts: TN, FP, FN and TP.

4.1 What accuracy hides

Key point: Accuracy says how many mistakes there are, not which kind.

A model with 90% accuracy makes mistakes 10% of the time. But in a two-class problem there are two kinds of mistake:

- a patient **has** heart disease, but the model says no;
- a patient does **not** have heart disease, but the model says yes.

These have very different consequences, and accuracy cannot tell them apart. The **confusion matrix** can.

4.2 Reading it

Key point: Rows are the actual classes, columns the predicted ones. The diagonal holds the correct predictions.

Figure 1 shows the confusion matrices of both models on the heart-disease test set (61 patients).

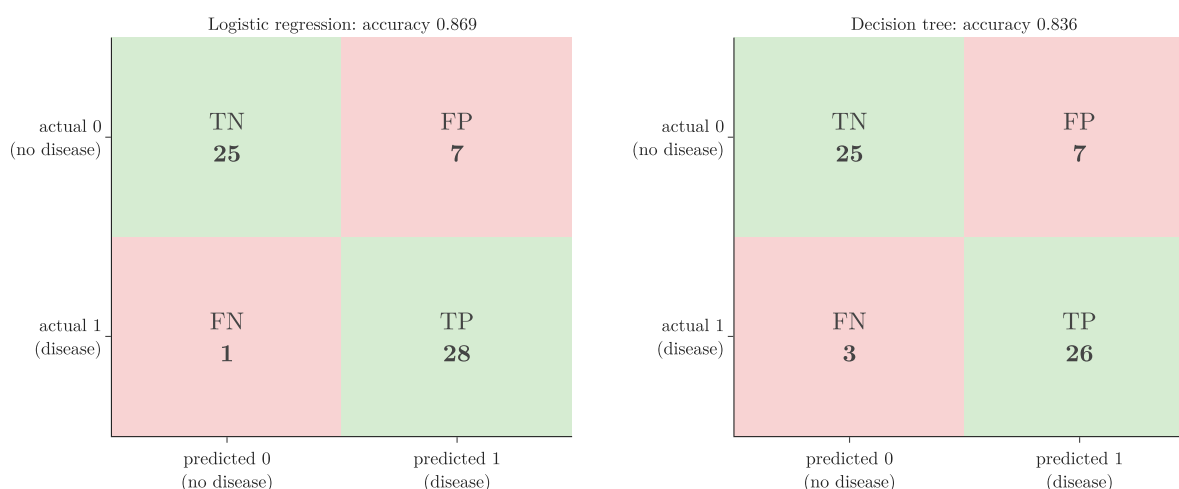


Figure 1: Confusion matrices for logistic regression and the decision tree

In scikit-learn's layout, each **row** is an actual class and each **column** a predicted class. For logistic regression:

	Predicted 0 (no disease)	Predicted 1 (disease)
Actual 0 (no disease)	25: true negatives (TN)	7: false positives (FP)
Actual 1 (disease)	1: false negatives (FN)	28: true positives (TP)

- **True positives (28):** patients with heart disease that the model correctly flagged.

- **True negatives (25):** healthy patients correctly cleared.
- **False positives (7):** healthy patients wrongly flagged as ill.
- **False negatives (1):** an ill patient the model missed.

The decision tree makes the same 7 false positives but 3 false negatives. So most of its extra errors are of the more dangerous kind: missed patients.

Extra: Some books and websites draw the matrix the other way round, with predictions as rows. Always check the axis labels before reading one.

4.3 The names

Key point: The second word is what the model predicted; the first word says whether that was right.

True / False Positive / Negative

first word: was the
prediction **right**?

second word: what
the model **predicted**

False Positive: predicted 1, wrong (really 0) = Type I error

False Negative: predicted 0, wrong (really 1) = Type II error

Figure 2: How the four names are built

- **Positive / Negative** is the model's prediction: 1 or 0.
- **True / False** says whether that prediction was correct.

So a “false positive” is a prediction of 1 that was wrong (the truth was 0). A “true negative” is a prediction of 0 that was right.

4.4 Type I and Type II errors

Key point: A false positive is a Type I error; a false negative is a Type II error.

These two error types also have names from statistics:

Error	Meaning	Heart example
Type I error = false positive	predicted 1, truly 0	a healthy patient told they have heart disease
Type II error = false negative	predicted 0, truly 1	an ill patient told they are healthy

4.5 Accuracy from the confusion matrix

Key point: Accuracy = $(TP + TN) / (TP + TN + FP + FN)$: the diagonal over the total.

The correct predictions are on the diagonal, so

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{28 + 25}{28 + 25 + 7 + 1} = \frac{53}{61} = 0.869$$

The confusion matrix gives the accuracy, but the accuracy cannot give back the confusion matrix: it carries less information.

Python: The confusion matrix in scikit-learn.

```
from sklearn.metrics import confusion_matrix
```

```
confusion_matrix(y_test, lr.predict(X_test))
```

```
# [[25  7]
```

```
# [ 1 28]]
```

```
tn, fp, fn, tp = confusion_matrix(y_test, y_pred).ravel()
```

5. More than two classes

Key point: With k classes the confusion matrix is $k \times k$. The diagonal is still the correct predictions; every other cell is a specific confusion.

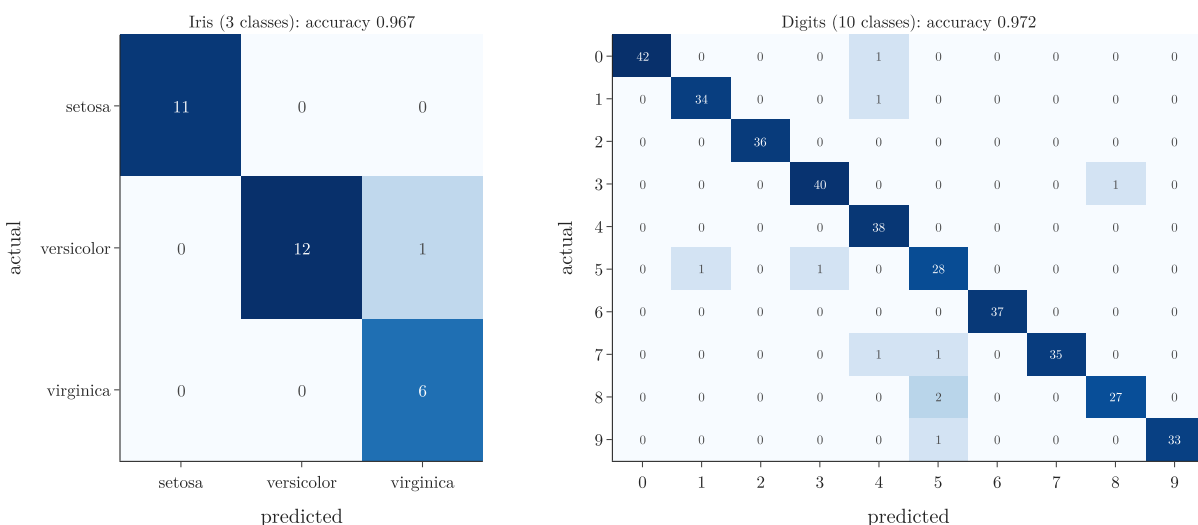


Figure 3: Confusion matrices for iris (3 classes) and digits (10 classes)

- **Iris** (left): 3×3 . The one mistake is a versicolor flower predicted as virginica (row versicolor, column virginica).

- **Digits** (right): 10×10 , from scikit-learn’s 8×8 pixel images of handwritten digits. Logistic regression gets 97.2% right. The off-diagonal cells show exactly which digits get confused: for example, two 8s were read as 5s.

Accuracy is again the diagonal total divided by the grand total.

6. When accuracy misleads: imbalanced data

Key point: If one class is very rare, a model that always predicts the common class gets very high accuracy while being useless.

Data is **imbalanced** when one class is much rarer than the other.

Imagine a model that screens airline passengers for security threats. Out of 100,000 passengers, perhaps 10 are real threats. A “model” that simply says “not a threat” for everyone gets

	Predicted 0	Predicted 1
Actual 0	99,990	0
Actual 1	10	0

$$\text{accuracy} = \frac{99,990}{100,000} = 0.9999$$

An accuracy of 99.99%, for a model that catches none of the 10 threats. The confusion matrix shows the problem at once: all 10 positives are false negatives.

So on imbalanced data, accuracy alone is the wrong metric. The next Note introduces **precision** and **recall**, which focus on the rare class.

7. Summary

Term	Meaning	Heart, logistic regression
TP	predicted 1, truly 1	28
TN	predicted 0, truly 0	25
FP (Type I)	predicted 1, truly 0	7
FN (Type II)	predicted 0, truly 1	1
Accuracy	$(TP + TN) / \text{total}$	0.869

- Accuracy is simple and works for any number of classes, but “good enough” depends on the problem.
- The confusion matrix shows the kind of each mistake; rows are actual, columns predicted in scikit-learn.
- On imbalanced data, accuracy can be near 100% for a useless model.

8. Key terms

Term	Meaning
Classification metric	A number that measures how well a classification model performs
Accuracy	The fraction of predictions that are correct
Confusion matrix	A table counting predictions for every pair of actual and predicted class
True positive (TP)	Predicted positive, and actually positive
True negative (TN)	Predicted negative, and actually negative
False positive (FP)	Predicted positive, but actually negative; a Type I error
False negative (FN)	Predicted negative, but actually positive; a Type II error
Imbalanced data	Data in which one class is much rarer than another